

```

#include<stdio.h>

#include<string.h>

void crc(char message[], char pattern[])
{
    int msg_length = strlen(message);
    int pattern_length = strlen(pattern);
    int fcs_length = pattern_length - 1;
    char msg[20];
    strcpy(msg,message);

    //Performing CRC division
    int i;
    for ( i = 0; i < msg_length; i++)
    {
        if (message[i] == '1' )
        {
            if(i<=msg_length-pattern_length)
            {
                for (int j = 0; j < pattern_length; j++)
                {
                    if (message[i+j] == pattern[j])
                        message[i+j] = '0';
                    else
                        message[i+j] = '1';
                }
            }
        }
    }
}

```

```
else break;
```

```
}
```

```
}
```

```
printf("Remainder R is : %s\n",message);
```

```
for (; i < msg_length + fcs_length; i++)
```

```
{
```

```
msg[i]=message[i];
```

```
}
```

```
printf("Message to be Transmitted T is : %s\n",msg);
```

```
char choice;
```

```
int position;
```

```
printf("\nDo You Want to Introduce error(Y/N): ");
```

```
scanf(" %c", &choice);
```

```
if (choice == 'Y' || choice == 'y')
```

```
{
```

```
x:
```

```
printf("Enter the Position: ");
```

```
scanf("%d", &position);
```

```
if (position < 1 || position > strlen(msg))
```

```
{
```

```
printf("Invalid position. Please enter a position between 1 and %d.\n", strlen(msg));
```

```
goto x;
```

```
}
```

```

//Introducing error at the specified position

msg[msg_length - position] = (msg[msg_length - position] == '1') ? '0' : '1';

}


printf("*** RECEIVER ***");

printf("\nMessage received at the Receiver :%s\n", msg);

for ( i = 0; i < msg_length; i++)
{
    if (msg[i] == '1' )
    {
        if(i<=msg_length-pattern_length)
        {
            for (int j = 0; j < pattern_length; j++)
            {
                if (msg[i+j] == pattern[j])
                msg[i+j] = '0';
            }
            else
            msg[i+j] = '1';
        }
        else break;
    }
}

printf("Remainder R is : %s\n",msg);

int flag=0;

```

```
for(i=0;i<msg_length;i++)
```

```
{
```

```
if(msg[i]=='1')
```

```
{
```

```
flag=1;
```

```
break;
```

```
}
```

```
}
```

```
if (flag)
```

```
{
```

```
printf("ERROR IN MESSAGE");
```

```
}
```

```
else
```

```
printf("MESSAGE WITHOUT ERROR");
```

```
}
```

```
int main()
```

```
{
```

```
char message[100];
```

```
char pattern[100];
```

```
char choice;
```

```
int position;
```

```
printf("*** SENDER ***");
```

```
printf("\nEnter Message: ");
```

```
scanf("%s", message);
```

```
printf("Enter Pattern: ");
```

```
scanf("%s", pattern);
```

```
int msg_length = strlen(message);
```

```
int pattern_length = strlen(pattern);
```

```
if (pattern_length >= msg_length)
```

```
{
```

```
printf("Pattern size should be less than Message. Please re-enter the Pattern \n");
```

```
return 0;
```

```
}
```

```
printf("Given Message is : %s\n", message);
```

```
printf("Given Pattern is : %s\n", pattern);
```

```
printf("Message size is : %d\n", msg_length);
```

```
printf("Pattern Size is : %d\n", pattern_length);
```

```
printf("FCS : %d bit\n", pattern_length - 1);
```

```
//Appending zeros to the message
```

```
for (int i = 0; i < pattern_length - 1; i++)
```

```
{
```

```
message[msg_length + i] = '0';
```

```
}
```

```
message[msg_length + pattern_length - 1] = '\0';
```

```
printf("After Appending Q is: %s\n", message);  
crc(message, pattern);  
}
```